

## SPECTRA SCHEDULER

Cognitive spectrum management  
ML-driven receiver scheduling — proposed

**Problem Statement ID: SIH26055**

Problem Statement Title

**Smart Scan strategy for  
Electronic Warfare**

PS Category: Software | Organisation: DRDO

Theme: [Confirm exact portal category]

Team ID: \_\_\_\_\_

Team Name (registered): \_\_\_\_\_



## Spectra Scheduler: confidence-aware adaptive scanning

Proposed solution: closed-loop spectrum scheduling that combines online learning, pulse association and receiver constraints to optimise frequency–time allocation.

### Detailed explanation of the proposed solution

Stochastic RF environment model.  
Pulse descriptor association.  
Contextual learning from feedback.  
Adaptive band and dwell selection.

### How it addresses the problem

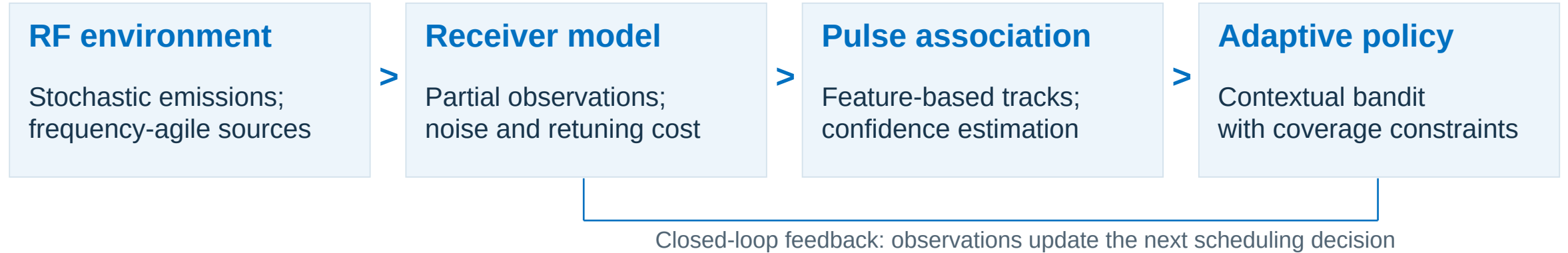
Reduce time to first interception.  
Adapt to non-stationary activity.  
Limit spectrum coverage gaps.  
Account for retuning overhead.

### Innovation and uniqueness of the solution

Confidence-aware scheduling.  
Joint discovery and revisit control.  
Change-responsive track memory.  
Traceable policy evaluation.

**DESIGN OBJECTIVE: HIGHER INTERCEPTION, LOWER SEARCH LATENCY**

## Methodology and process for implementation



## Technologies to be used

Python / NumPy: modular simulation and evaluation.  
HDF5 / h5py: planned bounded-memory ingestion.  
scikit-learn / PyTorch: proposed learning stack.  
Rust / PyO3: selective acceleration after profiling.

## Data and learning plan

TSRD features: ToA, frequency, width, angle, amplitude.  
Pipeline: schema validation > clustering > policy training.  
Learn from simulator hits/misses; hold out scenarios.  
Use file-level train / validation / test separation.  
Keep emitter identities outside inference features.

Proposed end-to-end architecture; simulation and heuristic evaluation form the existing foundation. [1–3]

## Technical feasibility and validation methodology

### Functional verification

Unit and integration testing  
Schema and regression checks

### Statistical evaluation

Seeded Monte Carlo experiments  
Paired baseline comparisons

### Planned stress testing

Noise and distribution-shift tests  
Latency and memory profiling

Engineering foundation: modular receiver simulation, deterministic replay and parallel evaluation.

| Potential challenges and risks               | Strategies for overcoming these challenges                       |
|----------------------------------------------|------------------------------------------------------------------|
| Dataset scale and memory pressure            | Chunked HDF5 ingestion; profiling-led native acceleration.       |
| Ambiguous pulse signatures and fragmentation | Multi-feature association; pairwise F1 and V-measure analysis.   |
| Concept drift and simulator overfitting      | Held-out scenarios, sensitivity analysis and policy ablations.   |
| Observation bias in recorded scan data       | Separate offline association from closed-loop policy evaluation. |

**Delivery plan:** data pipeline > association benchmark > learned scheduler > optimisation > API / dashboard.  
**Validation gate:** held-out gains over the strongest baseline, bounded memory and measured decision latency.

## Potential impact on the target audience

For electronic-support R&D teams, receiver-system developers and RF research laboratories:  
a software-defined evaluation platform for adaptive spectrum-management strategies.

### Technical benefits

Lower time to first interception.  
Improve interception probability.  
Preserve wide-spectrum coverage.

### Economic benefits

Reduce reliance on RF testbeds.  
Accelerate policy benchmarking.  
Reuse compute and test scenarios.

### Planned capabilities

Scalable offline data processing.  
Modular receiver integration.  
Analyst-facing visual diagnostics.

## Performance objectives and acceptance criteria

Target: improve the interception–latency trade-off against the strongest baseline at equal receiver budgets.  
Measure detection / false-alarm probability, sensitivity, interception rate, reward/cost and prediction accuracy.  
Track intercept-time error, coverage gaps, association F1 / V-measure, peak memory and decision latency.

## Details / links of the reference and research work

- 1 Gunn et al. — The Turing Synthetic Radar Dataset**  
Benchmark design, scan/stare receiver modes and pulse-association evaluation.  
[arxiv.org/abs/2602.03856](https://arxiv.org/abs/2602.03856)
- 2 Alan Turing Institute — TSRD dataset and companion code**  
Five pulse features; file-level splits; HDF5 loading and reference evaluation.  
[Hugging Face dataset](#) | [Turing Deinterleaving Challenge](#)
- 3 Gunn et al. — Transformer-based deep metric learning**  
Research basis for feature embeddings and context-aware pulse association.  
[arxiv.org/abs/2503.13476](https://arxiv.org/abs/2503.13476)
- 4 SIH26055 — Smart Scan strategy for Electronic Warfare**  
Problem requirements: receiver simulation, feedback learning and interception metrics.  
[SIH portal](#) | [Accessible statement copy \(community mirror\)](#)

Research foundation: reproducible benchmarking, contextual learning and multi-feature pulse association.